Reconstructing 3D Objects from Brain Activity
Final report | Iteration 4 | 11 October 2026

Problem and data

The task maps fMRI recordings to canonical 2048-point 3D object clouds. The supplied training split has 533 scans and 292 unique objects; validation has 125 scans and 67 objects; test has 167 scans and IDs only. Training fMRI was averaged over 10 time points for the retrieval baseline. The selected technical fallback uses only training object point clouds and the test IDs. Outputs preserve the supplied origin-centered, z-up coordinate frame.

Model selection and validation

The generic training medoid was strongest overall. Official evaluate.py metrics on 125 validation scans:

Chamfer 0.1562 | F-score 0.3947 | silhouette IoU 0.4384
silhouette SSIM 0.6919 | identification 48.76% | composite -0.013

Direct standardized-fMRI retrieval scored composite -0.158. The Iteration 2 predicted-shape-codebook model scored Chamfer 0.2211, F-score 0.2659, IoU 0.2939, SSIM 0.6844, identification 52.59%, composite -0.141. Its shuffled-association control preserved validation IDs, permuted only predictions, and used the same official evaluator. It scored identification 57.59% and composite -0.055, better than the unshuffled model. This single shuffle is not a significance test; brain dependence is not established.

A fold-local subject-centering experiment used five object-grouped training folds; global vs subject-centered latent MSE was 5.2491 vs 5.2651. This approximately 0.3% regression is a negative exploratory result, not official geometry evaluation. Group IDs were treated as split-local indices. Review also found Iteration 2 PCA/scaling were fitted before CV folds, a limitation of its internal CV estimate; no held-out validation targets fit the model. No validation target was used to fit transforms or model weights.

Test outputs and reproducibility

submission/predictions.npz contains 167 unique test IDs in original order and float32 points with shape (167, 2048, 3). Coordinates are finite. The same verified training-medoid cloud is repeated for all test scans. Therefore this fallback ignores test fMRI and is not a brain-conditioned solution; it may fail the assignment brain-use gate. No test ground truth was accessed. Calibrated PNG renders and PLY clouds are produced by the supplied starter helpers using cameras 0, 8, 16, and 24. No pretrained model was used.

Reproduction prerequisites and working directory

Run these commands in the assignment workspace root: the directory containing data/, starter.py, metrics.py, evaluate.py, and check_submission.py. Use Python 3.13 with NumPy, SciPy, and Pillow (install with python -m pip install -r submission/code/requirements.txt). The generator reuses the verified Iteration 3 archive when available; otherwise it uses the bundled training-derived generic_medoid.npy. The validation script invokes the supplied

official evaluator on validation targets; no test targets are used.

```
python submission/code/generate_submission.py
python submission/code/evaluate_validation.py
python check_submission.py submission
```

The official checker passed: predictions.npz, 167 PLY files, 668 renders, validation metrics, code, and report were verified.

Limitations and next steps

No tested model beats the generic medoid composite and shuffled control results do not support reliable scan-specific reconstruction. The submission is a technical backup, not a solved fMRI-to-3D model. The official submission checker passed for the listed artifacts, but that structural pass does not validate brain dependence. Any follow-on work should improve train-only brain-to-shape learning and use repeated permutation controls.

References

Gao et al., MinD-3D: Reconstruct High-quality 3D objects in Human Brain, arXiv:2312.07485 (2023).

Gao et al., MinD-3D++: Advancing fMRI-Based 3D Reconstruction with High-Quality Textured Mesh Generation and a Comprehensive Dataset, arXiv:2409.11315.